Hand-in sheet 2

Guidelines

- This hand-in sheet will be graded. It will count for 20% of the total grade. Please hand-in your answers before Noon Friday 12 June 2026.
- Answers may be handed in physically (Cc11) or digitally through e-mail to: jose.ezquiaga@nbi.ku.dk.
- You are allowed (and encouraged) to work in groups to discuss the exercises. However, everyone must write and hand-in their own answers (Show your work!).
- You are also allowed to use computer algebra software (e.g. Mathematica) and coding in general (e.g. Jupyter notebook or Python) to aid with your computations. However, if you do, please include your notebooks with your answer.
- Please make sure that your answers (and any supplemental material such as code – if needed) are comprehensible!

Exercise 1: Radiative and non-radiative degrees of freedom

On a global vacuum spacetime we have seen that the equations of motion only contain two physical propagating degrees of freedom that follow a wave equation. These are *radiative* modes. On a global spacetime with matter sources, however, this is not generically true. There could be physical, *non-radiative* degrees of freedom. These modes are characterized by Poisson-type equations. Schematically, instead of $g_{\alpha\beta}\nabla^\alpha\nabla^\beta\phi_A = \cdots$ one has $\gamma_{ij}\nabla^i\nabla^j\phi_A = \cdots$, where γ_{ij} is the spatial metric.

Let's begin by decomposing the metric perturbations $h_{\mu\nu}$, which is considered to be a linear perturbation over flat space-time,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}\,, \qquad ||h_{\mu\nu}|| \ll 1\,,$$

into irreducible representations of the rotation group:

$$\begin{aligned} h_{00} &= -2\Phi\,, \\ h_{0i} &= w_i\,, \\ h_{ij} &= 2s_{ij} - 2\Psi\delta_{ij}\,, \end{aligned}$$

where Φ is a scalar, w_i a three-vector, s_{ij} a symmetric and traceless three-tensor and Ψ is a scalar encoding the trace

$$\begin{aligned} \Psi &= -\frac{1}{6}\delta^{ij}h_{ij}\,, \\ s_{ij} &= \frac{1}{2}\left(h_{ij} - \frac{1}{3}\delta^{kl}h_{kl}\delta_{ij}\right)\,. \end{aligned}$$

δ_{ij} is the spatial Kronecker's delta.

Following this decomposition:

1. Show how each of the metric components (Φ, w_i, Ψ and s_{ij}) transform under gauge transformations $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$.

2. Show that it is possible to choose a gauge, "transverse gauge", in which $\partial_i s^{ij} = 0$ and $\partial_i w^i = 0$.

3. Compute the Einstein field equations in the *transverse gauge* for the linear metric perturbations and a generic energy-momentum tensor $T_{\mu\nu}$.

4. Show that only s_{ij} follows a wave-equation.

If you are looking for inspiration, this is nicely discussed in Flanagan and Hughes review [1] as well as Carroll's book [2].

Exercise 2: Quasi-monochromatic gravitational wave sources

During the course we have studied the gravitational waves (GWs) produced during the inspiral phase of a compact binary coalescence. At leading order, the amplitude of the GW scales as:

$$h(t) = \frac{4}{d_L}\left(\frac{G\mathcal{M}_z}{c^2}\right)^{5/3}\left(\frac{\pi f_{\rm gw}(t)}{c}\right)^{2/3}\cos(\Phi(t))\,,$$

where d_L is the luminosity distance, c the speed of light and G Newton's constant. Note that we have omitted here the polarization dependence of the strain on the inclination angle. The phase is determined by the frequency evolution

$$\Phi(t) = 2\pi \int dt f_{\rm gw}(t)\,.$$

Recall that $\mathcal{M}_z = (1+z)\mathcal{M}_c$ is the redshifted chirp mass and $f_{\rm gw}(t)$ the frequency evolution measured at the detector, which can be computed through the energy loss of the orbit in GWs:

$$\dot{f}_{\rm gw} = \frac{96}{5}\pi^{8/3}\left(\frac{G\mathcal{M}_z}{c^3}\right)^{5/3} f_{\rm gw}^{11/3}\,.$$

Depending on the mass of the binary and the frequency range of observation, a given signal can be in different regimes. When for a given observing time $T_{\rm obs}$ the frequency evolves little compared to the signal itself,

$$\dot{f} \cdot T_{\rm obs} \ll f\,,$$

the signal is said to be *quasi-monochromatic*.

1. Show that for a quasi-monochromatic source, the GW signal can be written at leading order as
$$h(t) = A \cos\left[2\pi\left(ft + \frac{1}{2}\dot{f}t^2\right) + \phi_0\right],$$
where A and ϕ_0 are constants.

   For a binary neutron star of component masses equal to $1.4M_\odot$ that is observed for a year, what is the frequency range in which the quasi-monochromatic approximation will be valid? How does this condition scale with the time of observation and chirp mass?

2. The number of sources emitting GWs with a frequency larger than f is given by
$$N_{\rm gw}(> f) \sim R V_c \Delta t_{\rm merge}(f)\,, \tag{0.0.1}$$
where R is the comoving merger rate, V_c is the comoving volume and $\Delta t_{\rm merge}$ the time to merger for a signal with initial frequency f.

   How long it takes to merge a binary neutron star with an initial frequency of 1mHz? How does the number of events scale with the initial frequency?

   Given that from the latest GW catalog of LIGO–Virgo–KAGRA we know that the comoving merger rate of binary neutron stars is within 10–1700$\mathrm{Gpc}^{-3}\mathrm{yr}^{-1}$, and that the number density of Milky-way like galaxies is about $n_{\rm MW} \sim 0.01\mathrm{Mpc}^{-3}$, what would be the number of sources that we should expect to be emitting GW with frequencies larger than 1mHz in our own galaxy?

3. We want to compute the signal-to-noise ratio (SNR) of a quasi-monochromatic signal. To that end, first demonstrate that the noise-weighted inner product $(a|b)$ of two real functions $a(t)$ and $b(t)$ defined as
$$(a|b) = 4\mathrm{Re}\int_0^\infty \frac{\tilde{a}^*(f)\tilde{b}(f)}{S_n(f)}df$$
can be equivalently written in time domain as
$$(a|b) = \int_{-\infty}^{\infty} dt\hat{a}(t)\hat{b}(t)\,,$$
where $\hat{a}(t)$ is the inverse Fourier transform of the frequency domain noise-weighted signal $\tilde{\hat{a}} = \tilde{a}(f)/\sqrt{S_n(f)/2}$.

   Then, show that for a quasi-monochromatic, the SNR accumulated during a time of observation $T_{\rm obs}$ at a frequency f is
$$\rho = \sqrt{(h|h)} \simeq \frac{A}{\sqrt{S_n(f)}}T_{\rm obs}^{1/2}\,.$$

Bibliography

[1] E. E. Flanagan and S. A. Hughes, *The Basics of gravitational wave theory*, *New J. Phys.* **7** (2005) 204, [`gr-qc/0501041`].

[2] S. M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity*. Cambridge University Press, 7, 2019. Based on gr-qc/9712019.